

PRAKTIKUM 2 REPORT

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Wien, 2026

Studienkennzahl lt. Studienblatt: UA 066 921

Fachrichtung: Informatik – Allgemein

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Abstract

The purpose of this report is to keep its readers up to date with the second update of my Bachelor thesis project. It directly builds upon what was done in Practical course 1. The upgrade includes grammar correction for the input, sentiment analysis, aspect-based sentiment analysis, hate and abuse detection, semantic role labelling, and finally the replacement of the old model with the new Ollama one. Grammar correction aims at finding an error in the user's input text and then suggesting an improvement. Sentiment analysis grades the text as positive, negative, or neutral. The aspect-based sentiment analysis is similar; however, it also tries to connect the words that prove why it determined the outcome. Hate and abuse detection aims to categorize the input sentence into a specific group type, depending on the type of discrimination. Semantic role labelling shows who did what to whom, when, where, and how in a sentence. This upgrade was done in the existing Flask project, with the aim of further improving the web-app as a whole, providing a more complete tool for the analysis of the Serbian language. Apart from that, everything done here will be used as a basis for the master's thesis.

1 Introduction

In recent years, automatic processing of texts with Large Language Models (LLMs) has become increasingly popular. This interest has also pushed the scientists and developers to further explore the capabilities and limitations of such processing systems. LLMs are able to better capture different types of information from text, such as contextual, syntactic and semantic information, compared to traditional machine learning approaches. Exactly this capability is why they are used for grammar correction, sentiment analysis, semantic understanding, content detection, and moderation across multiple domains [16, 2].

Despite all of this progress, the majority of LLM tools and research are still focused on high-resource languages, with English in the center. Languages with freer word order and richer morphology, such as Serbian, are still considered undertrained. Lack of training data combined with higher complexity is the main reason for this lack of robustness. However, this also means that the Serbian language is a valuable test case for evaluating the performance and adaptability of LLMs in more complex linguistic settings. Extending LLM-based analysis of Serbian contributes both to practical applications and to the research in broader multilingual natural language processing[15].

This report is the second update of a Bachelor thesis project focused on providing a central and unified way to locally execute Serbian language analysis. It builds upon a web-based application using the Flask micro web framework as a modular extension of the Practical Course 1, with the aim of significantly extending the analytical scope. The main objective of the update is to improve the semantic and pragmatic depth of the text analysis by integrating the Llama3.1:8b model.

The first of the features introduced is grammar correction, which detects the error and suggests an alternative corrected version. As the name suggests, this aims to detect inconsistencies and then produce a sufficiently good alternative. In addition to this, sentiment analysis has been introduced, providing users the ability to classify their text as coarsely-grained positive, negative, or neutral tones[11]. This functionality is then extended in aspect-based sentiment analysis, which attempts to associate the sentiment polarity with specific aspects or terms in the sentence that support it, enabling a more fine-grained analysis[12].

Furthermore, the system now also supports hate and abuse detection, a task that has gained importance due to the harmful and discriminatory content produced daily online. This extension categorizes input based on different types of abuse, which contributes to safer and more responsible language applications[17]. Additionally, semantic role labelling has been added. This supports deep semantic analysis by identifying the predicate-argument structures that answer the question of who did what to whom, when, where and how[6].

The final extension to the project is the replacement of old models with the newest Llama3.1:8b model deployed within Ollama framework. This change was made due to better performance on user input compared to other models. All of the enhancements mentioned in this report will serve as a foundation for the future work in the context of the upcoming master’s thesis.

2 Grammar Correction and Suggestion

The task considered as a foundation of natural language processing is grammar correction. Its task is to identify and correct grammatical, orthographic, and agreement errors in the input while preserving the original meaning. This process originally relied on rule-based systems or statistical models, which worked well on simple sentence structures in less morphologically rich languages [8]. However, these models performed poorly when used on morphologically rich languages with highly complex sentence structures, something that was improved with recent advances in LLMs. The quality of grammar correction has improved significantly by enabling context-based and semantically informed error detection and correction [2, 16].

Within this project, a prompting-based approach has been used with an LLM deployed locally through the Ollama framework to implement grammar correction. The entire correction process is driven by specifically designed prompts. The model is prompted to act as a Serbian proofreader using the Ekavian dialect to correct the input sentence. It is also instructed to keep the exact word order and not to change the meaning of the sentence, as long as it is grammatically correct. The final prompt states that only the sentence should be returned, without any explanations of what was done or corrected. These prompts were chosen because they allow the system to handle a wide range of grammatical phenomena, such as agreement, word order, and orthographic consistency. These qualities are especially challenging to detect in the Serbian language due to its relatively free word-order and rich morphology [15]. The constraints used in the prompts are important, since LLMs struggle with unnecessary paraphrasing, a common issue when using generative language models for such tasks [13].

One final factor that influences the output of the LLM is the temperature. What this parameter influences is how deterministic the model is, or in other words, it sets the randomness of the model. The higher it is, the more random it is. The lower it is, the more deterministic it is. Because the model is used for grammar correction, which requires consistency and precision rather than creativity [7]. In the project, a very low temperature of 0.05 was used to maintain a near-deterministic behaviour, while allowing minimal flexibility. The non-zero number can help avoid rare cases where the greedy algorithm locks into locally optimal but globally suboptimal continuation [7].

Input Sentence: "Ja ideš da jedem u restoraan."

Mistakes in text: ideš - wrong person; restoraan - extra "a".

Expected corrected sentence: "Ja idem da jedem u restoran."

Translated correct sentence: "I got to eat at a restaurant."

Table 1: Effects of different temperature values in grammar correction.

Temperature	Correct	Model Output
0.0	Yes	Ja idem da jedem u restoran.
0.1	Yes	Ja idem da jedem u restoran.
0.2	No	Ja ideš da jedem u restoran.
0.3	Yes	Ja idem da jedem u restoran.
0.4	Yes	Ja idem da jedem u restoran.
0.5	Yes	Ja idem da jedem u restoran.
0.6	No	Ja ideš da idem u restoran.
0.7	No	Ja ideš da jedem u restoran.
0.8	No	Ja ideš u restoran da jedem.
0.9	No	Ja ćete ići da jedem u restauraciju.
1.0	Yes	Ja idem da jedem u restoran.

Even in simple sentences, higher temperatures usually do not produce a satisfactory result. Further testing with a fixed temperature value of 0.05 was conducted, where the following was tested: incorrectly written words, wrong word order, and subject-verb agreement errors. Test sentences had varying numbers of mistakes.

The results show that the model exhibits strong grammar correcting capabilities, particularly in correcting the word order, spelling mistakes, correcting verb forms, and multiple errors in one sentence. However, it was also revealed that, even with the near-deterministic settings, the model sometimes introduced unintended changes, like changing the word genus from male to female, using a different dialect, and adding or subtracting letters from otherwise correct sentences.

Overall, the LLM used offers a good assisting hand in correcting simple text snippets that have clear mistakes. However, it still requires human review and assistance. The prompts used do not guarantee the adherence to the task-specific constraints.

3 Sentiment Analysis

Apart from grammar correction, sentiment analysis is also a core task in natural language processing. It is responsible for identifying and classifying the emotion expressed in the input text. Most commonly, it classifies the emotion into three classes: positive, negative, and neutral [9]. As with grammar correction, early

approaches to sentiment analysis relied on rule-based methods and manually engineered features in machine learning models [11].

Sentiment analysis also improved with recent advances in deep learning and transformer-based architectures by enabling context-aware representations of text[16]. Even when trained on heterogeneous or weakly supervised data, models like multilingual transformer models show great generalization capabilities for multiple languages and across different domains [5].

This project implements sentiment analysis using a pre-trained multilingual model, **twitter-xlm-roberta-base-sentiment**. The model is run locally via Hugging Face Transformers. The architecture is based on XLM-ROBERTa, which was trained on a large corpus of Twitter data annotated specifically for sentiment [1]. The system first analyzes the input as a whole and then sentence by sentence.

Input Text: "Projekat radi veoma dobro. Juče sam radio na zadatku. Situacija je veoma frustrirajuća."

Translated Input: The project is working very well. I was working on the assignment yesterday. The situation is very frustrating.

Table 2: Sentiment Analysis on three inputs

Sentence	Negative	Neutral	Positive
Projekat radi veoma dobro.	0.021	0.064	0.915
Juče sam radio na zadatku.	0.114	0.598	0.288
Situacija je veoma frustrirajuća.	0.912	0.070	0.018

When tested on single sentences, the model mostly provides reliable and accurate results, especially when the sentiment is clearly expressed, which increases the confidence. On very rare occasions, the model hallucinated the output, in the sense that the sentiment was clearly negative and the input contained multiple negative words, but the model ranked it as positive with high confidence. One additional limitation is Recency Bias, which means that the model is going to determine the sentiment of the entire input based on the last part of the sentence. If the sentence is long and starts with something positive but ends with a negative expression, then negativity will dominate the confidence score [3]. Here is an example of how recency bias works:

Input: "Mrzim ljude ali volim lepe životinje."

Translated Input: I hate humans but I like beautiful animals.

Output: POSITIVE (conf=0.834)

Input: "Volim lepe ljude ali mrzim životinje"

Translated Input: I love beautiful humans but I hate animals.

Output: NEGATIVE (conf=0.892)

4 Hate and Abuse Detection

Hate and abuse detection can be seen as an extension of sentiment analysis, specifically describing what type of negative content has been detected. The goal of this task is to identify language that targets a specific individuals or groups based on characteristics such as religion, race, gender, sexual orientation, and disability [17]. Due to the nature of the task, specifically subjectivity and context, hate speech detection remains a challenging problem for low-resource languages [14].

The project implements hate and abuse detection using a pre-trained multilingual model, `sadjava/multilingual-hate-speech-xlm-roberta`. This transformer-based model is based on the XLM-RoBERTa architecture and shows strong detection capabilities for abusive language due to its ability to capture semantic relationships. For each input received, the model outputs confidence scores for each of the categories previously mentioned, as well as the *appropriate* class representing non-hateful content [4]. The classification decision is determined by a threshold-based policy applied to all output probabilities. If any of the confidence scores exceed the predefined threshold, the input is flagged accordingly.

Table 3 presents the main detection behaviour of the model. The results shown in the table and in other independent tests show that the model has a high chance of flagging an input, even though it is not considered a negative one. Constructing a sentence in which one expresses positive sentiment toward a specific nationality, religion, or sexual orientation results in the input being flagged under the corresponding category rather than as *appropriate*. Despite this observation, the model itself is still accurate in the sense that it does not confuse different types of discrimination. Finally, the model itself is very fast and runs in milliseconds when processing a single input. Overall, the model is mostly successful in capturing overtly harmful language patterns.

Input 1: Ubijati ljude nije u redu.

Translated Input 1: Killing people is not okay.

Input 2: Ovo je obična izjava o Jevrejima bez uvreda.

Translated Input 2: This is an ordinary statement about Jews without any insults.

Input 3: Mrzim Amerikance i ne želim da imaju ista prava.

Translated Input 3: I hate Americans and I do not want them to have the same rights.

Input 4: Treba transrodne osobe ukloniti iz društva.

Translated Input 4: Transgender people should be removed from society.

Input 5: Svi retardirani ljudi su isti.

Translated Input 5: All mentally disabled people are the same.

Table 3: Testing Hate and Abuse Detection

Input Sentence	Flagged	Predicted Category	Interpretation
1	No	appropriate	Correct detection
2	Yes	religion	False positive
3	Yes	race	Correct detection
4	Yes	sexual orientation	Correct detection
5	Yes	disability	Correct detection

5 Aspect-Based Sentiment Analysis

An extension of traditional sentiment analysis is aspect-based sentiment analysis (ABSA). Like sentiment analysis, it detects the sentiment polarity, but with respect to specific aspects or entities mentioned in the sentence. It does not assign a global sentiment to the whole sentence but to a pair of words found within the sentence, where one word is an aspect and the other evidence. Evidence is used as proof for the assigned sentiment. Such an approach enables a more fine-grained interpretation of the content, especially useful in product reviews and feedback. [12].

This project implements ABSA using a hybrid approach. It combines the syntactic information from the CLASSLA pipeline with semantic reasoning from the model *llama3.1:8b*. The CLASSLA pipeline performs tokenization, lemmatization, part-of-speech tagging, and dependency parsing on the input text. The extracted dependency relations are then used to extract aspects, focusing on nouns that play important grammatical roles in the sentence [15]. The aspects are then passed on to the LLM deployed locally via the Ollama framework. The model is specifically prompted to perform sentiment classification independently for each aspect and then assign one of the three sentiment labels along with confidence score [9]. The final step is post-processing the model output in order to improve consistency. A list of predefined cues, both positive and negative, was used to validate the sentiment output of the model. In case of contradictions, the polarity and confidence were adjusted accordingly. This hybrid approach has shown flexible results [11].

The functionality was evaluated on a set of manually constructed sentences that contained multiple aspects with the evidence. To make the task harder,

the sentences also contained potentially conflicting sentiments. The results show that aspects are successfully detected and separated. Sentences expressing both positive and negative opinions towards different aspects are also correctly detected and separated.

However, some limitations were also detected in this approach during testing. In special cases, abstract nouns or temporal expressions are identified as aspects and assigned labels, which should not be the case. This reflects the difficulty of aspect extraction in morphologically rich and free word order languages such as Serbian [15].

Input 1: Kamera je odlična, ali baterija je slaba.

Translated Input 1: The camera is excellent, but the battery is weak.

- **Aspect 1:** kamera (camera)
 - **Sentiment:** positive
 - **Confidence:** 0.90
 - **Evidence:** odlična (excellent)
- **Aspect 2:** baterija (battery)
 - **Sentiment:** negative
 - **Confidence:** 0.80
 - **Evidence:** slaba (weak)

Input 2: Ekran je pregledan i svetao, međutim cena je visoka.

Translated Input 2: The screen is clear and bright; however, the price is high.

- **Aspect 1:** ekran (screen)
 - **Sentiment:** positive
 - **Confidence:** 0.80
 - **Evidence:** pregledan (clear and bright)
- **Aspect 2:** cena (price)
 - **Sentiment:** negative
 - **Confidence:** 0.90
 - **Evidence:** visoka (high)

6 Semantic Role Labelling

The objective of Semantic Role Labelling (SRL) is to identify predicate-argument structure in a sentence. To be precise, SRL is concerned with "identifying who does what to whom in an event or state described by a predicate," and with

”various notions of when, where, and how.” SRL is closely related to syntactic parsing; however, SRL is concerned with predicate-argument structure rather than syntactic parsing. SRL is primarily used in higher-level language understanding and information extraction [6, 10].

SRL is carried out in the current project through a hybrid methodology that relies on both a large language model (LLM) [16, 2] and a linguistic pipeline in Serbian. The linguistic pipeline includes tokenization, lemmatization, part-of-speech tagging, and morphological analysis. The outputs of these operations are further used to create a token sequence with indices starting from 1 and to identify candidate predicates by focusing on tokens that carry one of the following tags: VERB or AUX, including verbal participles [15].

The model is then prompted to produce a strict JSON response that follows a given schema [5]. The JSON structure contains mappings of sentences to one or more frames. Each frame contains two dictionaries: one for the predicate and another for semantic roles. The set of semantic roles is kept very concise and intuitive. The set of roles includes Agent/Doer, Patient/Theme, Recipient/Beneficiary, Time, Location, Destination, etc. Finally, the JSON response is normalized through a set of rules. For example, temporal adverbs are mapped to Time, causal prepositions like ”zbog” to Cause, and Location is further specified to Source or Destination depending on the governing preposition. To address common SRL ambiguity issues, additional rules address control predicates and clitics [4].

Input: Marko je juče kupio knjigu u prodavnici.

Translated Input: Marko bought a book in the store yesterday.

Output (predicates): *kupio (bought)*

Agent/Doer: Marko

Patient/Theme: knjigu (book)

Time: juče (yesterday)

Input: Ana je dala Petru poklon zbog rođendana.

Translated Input: Ana gave Peter a gift because of his birthday.

Output (predicates): *dala (gave)*

Agent/Doer: Ana

Patient/Theme: poklon (present)

Recipient/Beneficiary: Petru

Cause: zbog rođendana (because of the birthday)

Input: Mi želimo da idemo u Beograd sutra.

Translated Input: We want to go to Belgrade tomorrow.

Output (predicates): *želimo (we want), idemo*

Agent/Doer: Mi (we)

Destination: u Beograd (in Belgrade)

Time: sutra (tomorrow)

The above examples illustrate the frames produced by the web-app. The words of the first sentence are correctly assigned: "Marko" is assigned as Agent/Doer, "knjiga" as Patient/Theme and "juče" as Time. The same accuracy applies to the second sentence. As for the third sentence, the system creates separate frames for control predicate "želim" and the embedded predicate "idem", propagating the Agent/Doer "Mi" and identifying "sutra" as Time.

Despite these examples, there are still limitations observed during the testing phase. Auxiliary verbs, such as the forms of "biti" are consistently extracted as separate predicates, which sometimes introduce redundant frames that are not always meaningful for the end users. Furthermore, some semantic roles are not always correctly assigned. Instrumental phrases may be labeled as Patient/Theme and complex prepositional phrases may result in broad role spans, like destinations including purpose clauses. These limitations highlight the challenges of precise predicate-argument boundary detection and role disambiguation, especially in morphologically rich languages such as Serbian [15, 4].

7 Appendix

7.1 Repository

The Github repository of the project can be found [here](#). Steps on installation and usage can be found in the README document.

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